



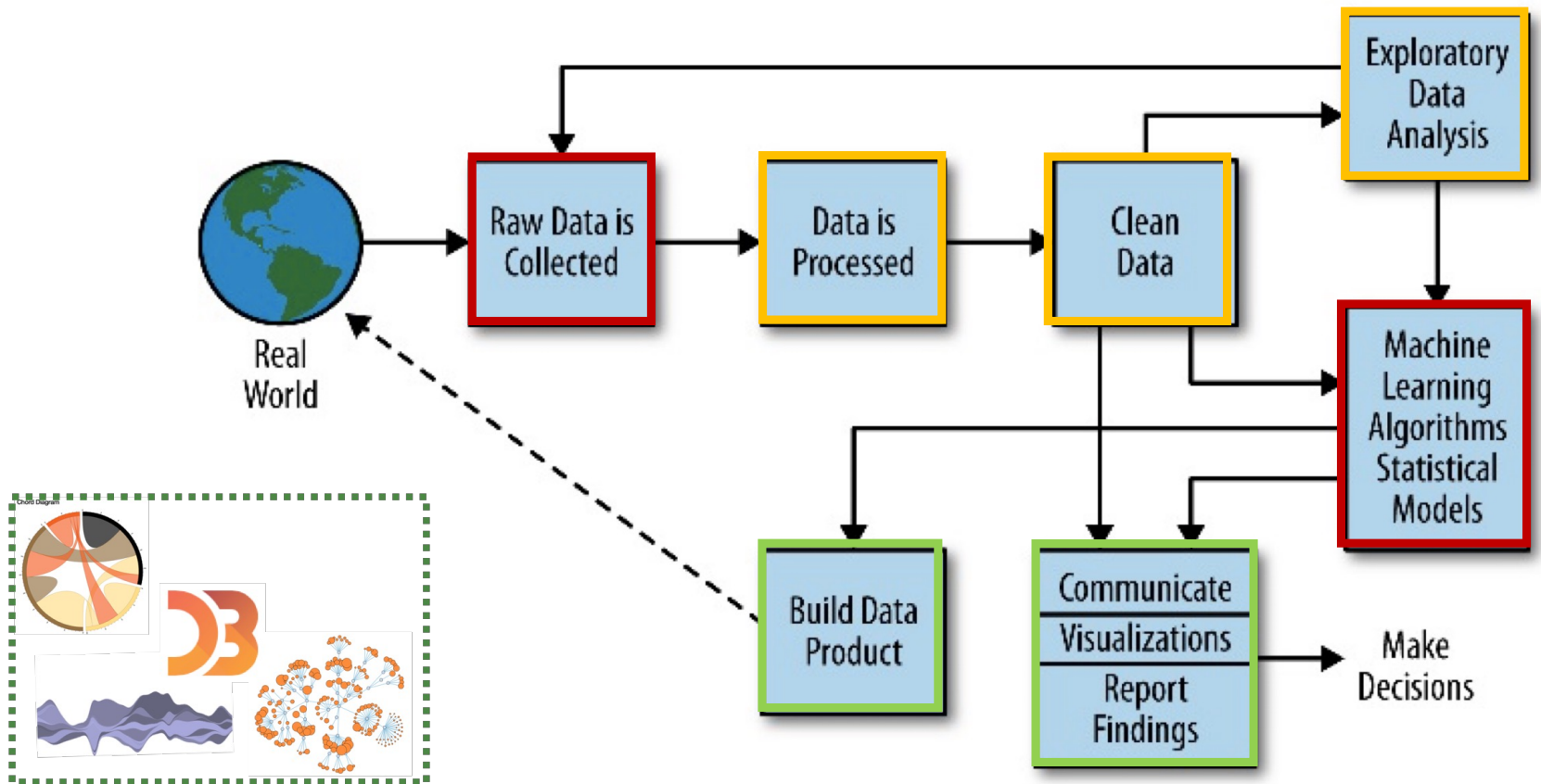
W1.1 COS30045

Lecture

Agenda

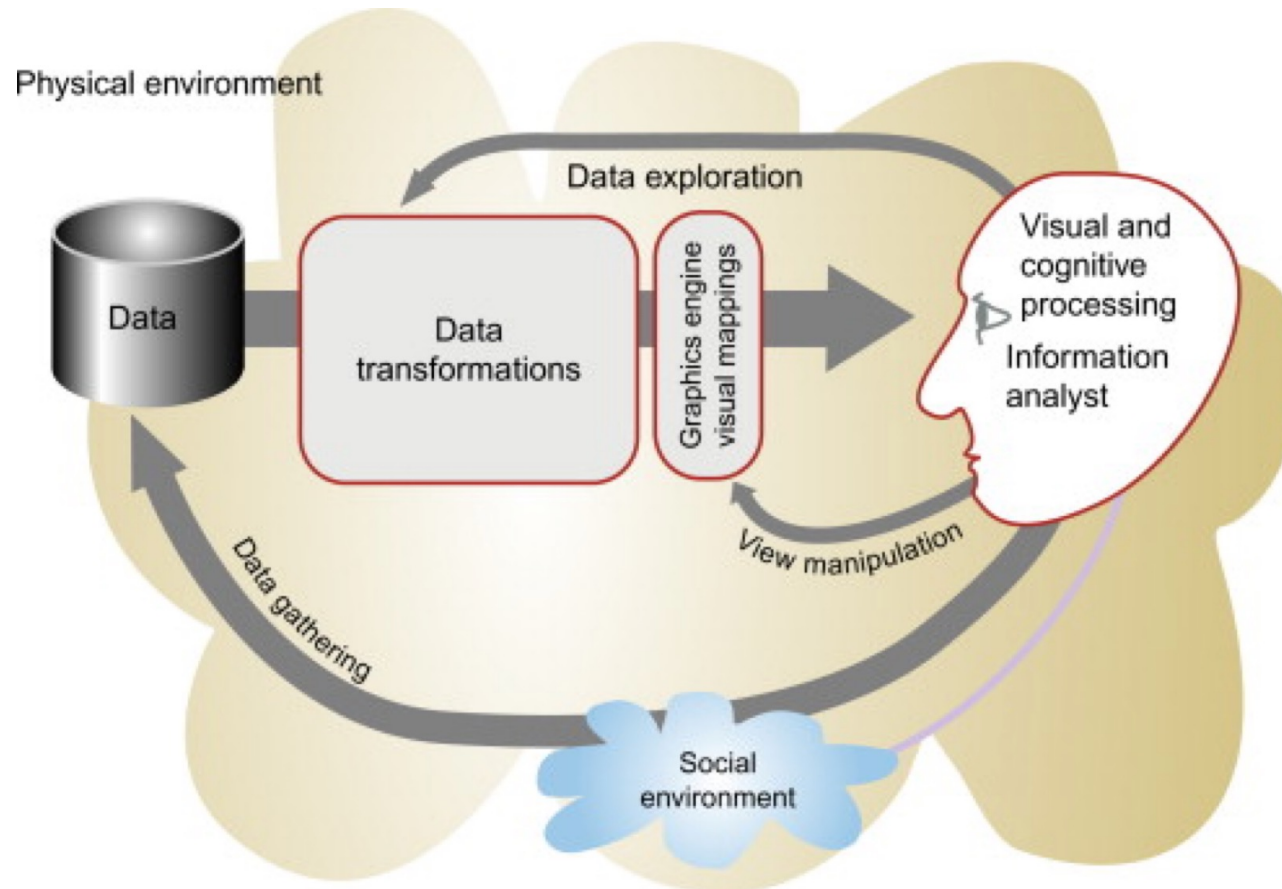
- Introduction to Data Visualisation

What this unit covers...



Schutt, R., & O'Neil, C. (2014). *Doing data science: Straight talk from the frontline*. O'Reilly.

What this unit is about...



Why study data visualisation?

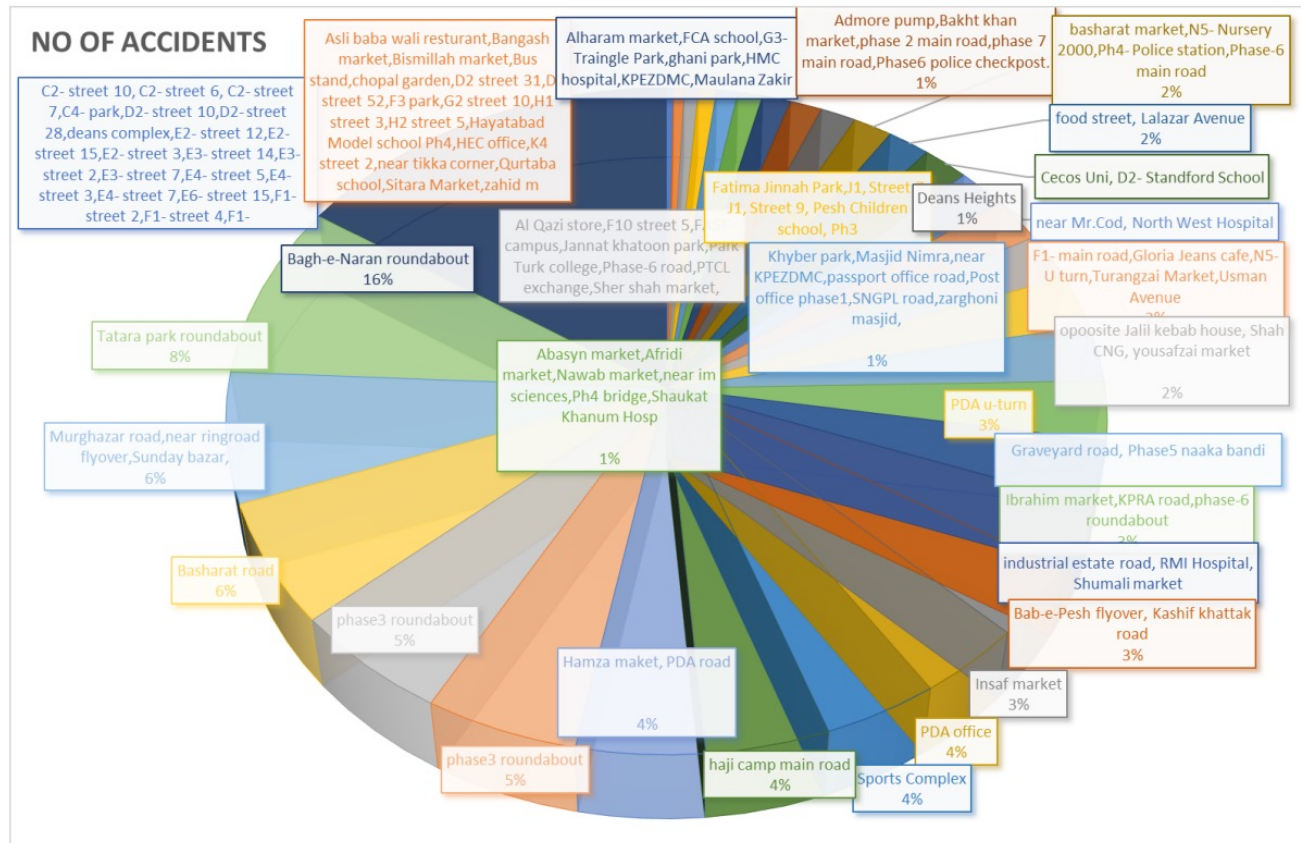
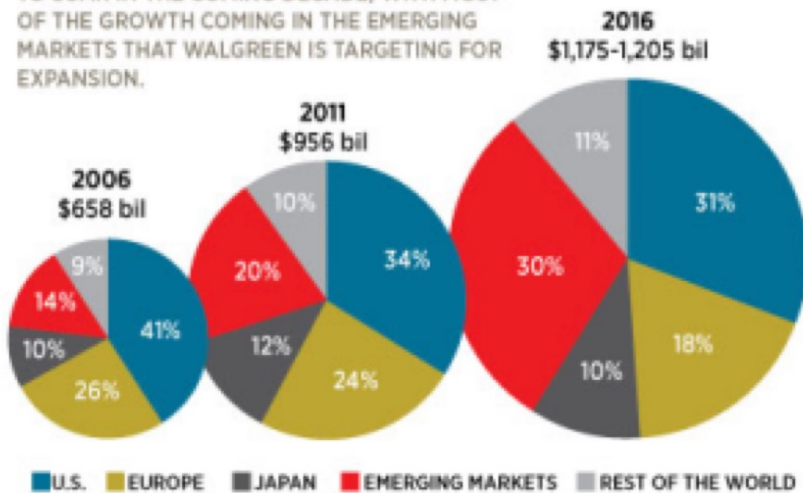


Figure 6. Accidents locations w.r.t percentage

Why study data visualisation?

A WORLD OF DRUGS

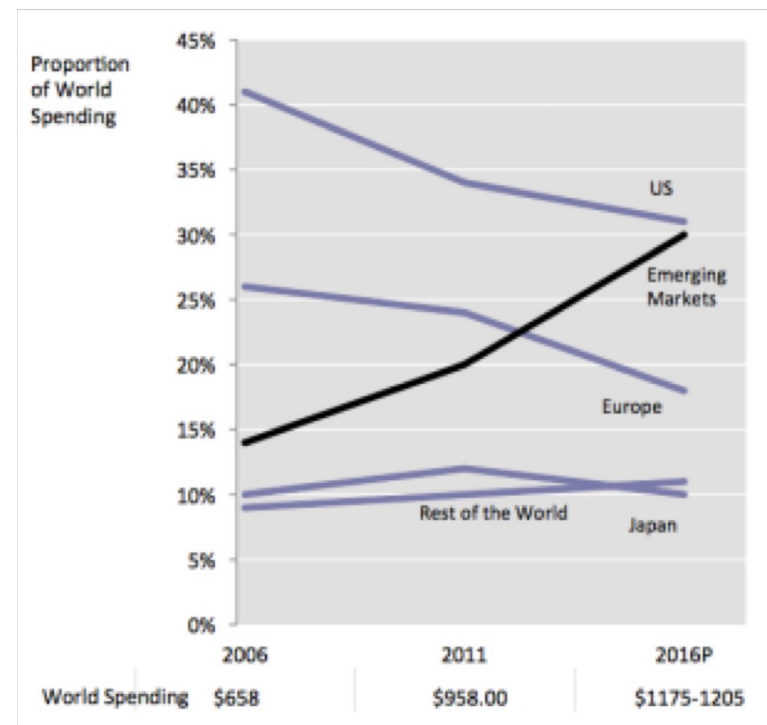
GLOBAL SPENDING ON MEDICINE IS EXPECTED TO SOAR IN THE COMING DECADE, WITH MOST OF THE GROWTH COMING IN THE EMERGING MARKETS THAT WALGREEN IS TARGETING FOR EXPANSION.



SOURCE: IMS INSTITUTE FOR HEALTHCARE INFORMATICS.

Which market is growing fastest?
Which market/s are declining?

[Junk Charts \(2012\) Expanding circles of error](#)



Valid?

Many more people in emerging markets and rest of world than in US and Japan...

Why study data visualisation?

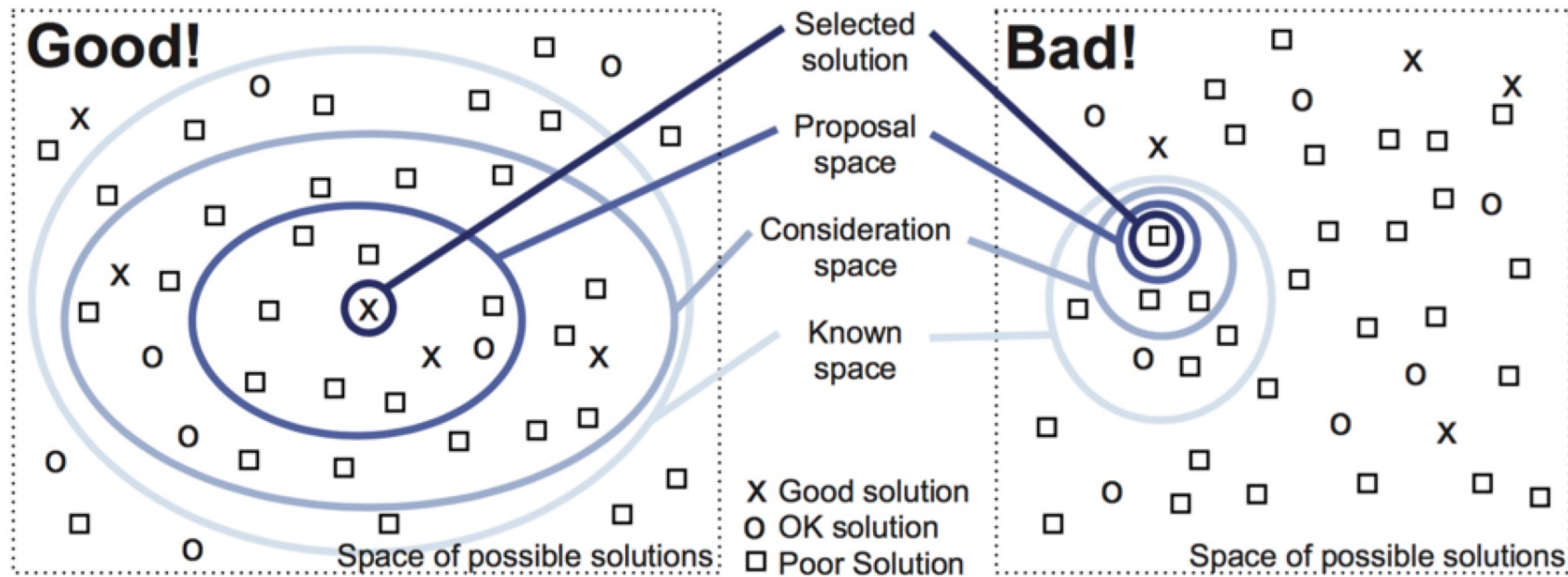
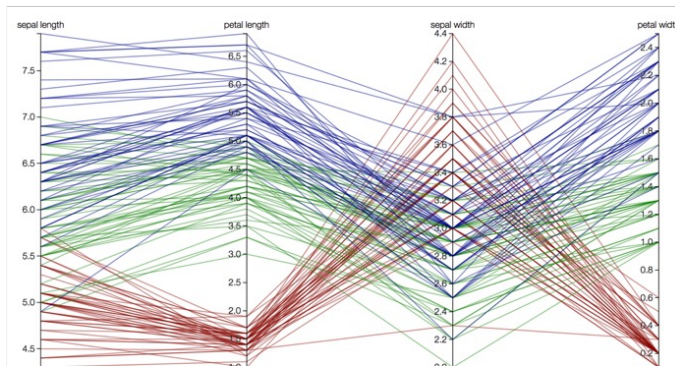


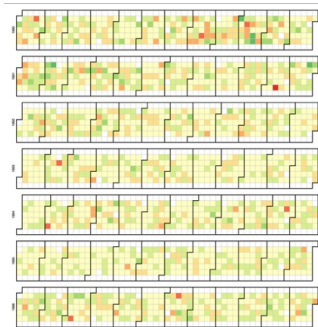
Figure 1.5. A search space metaphor for vis design.

Design space...

Parallel coordinates



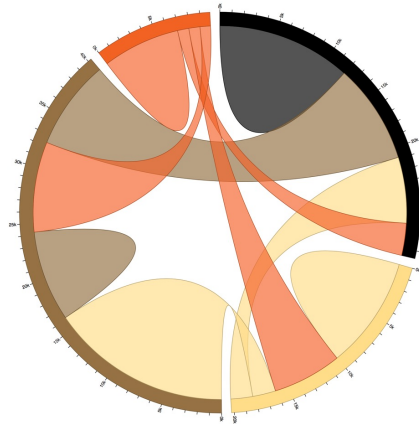
Calendar view



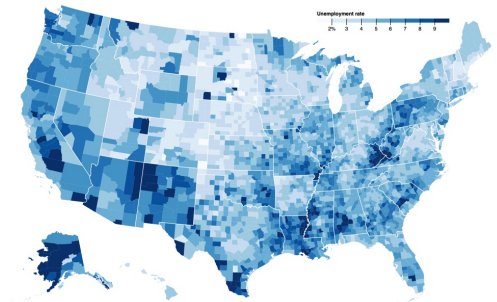
Tree map



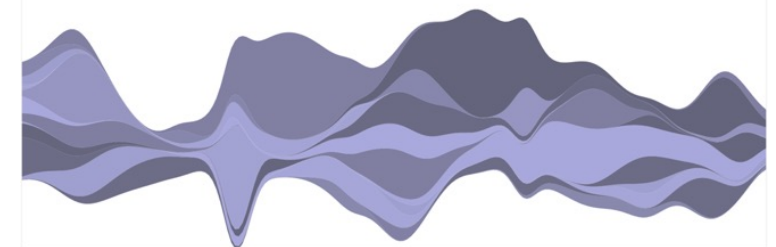
Chord Diagram



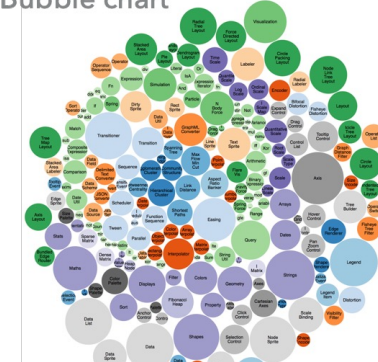
Choropleth



Streamgraph



Bubble chart



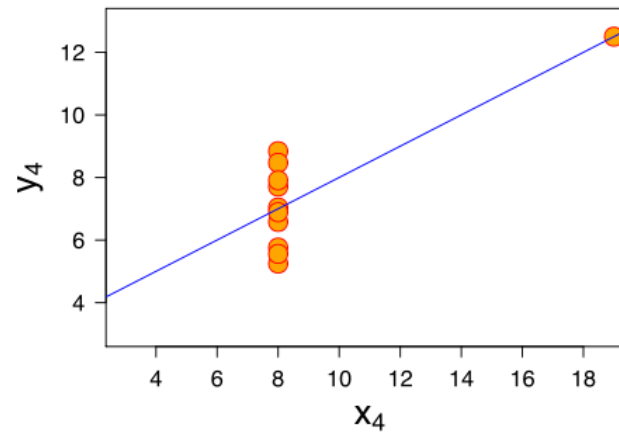
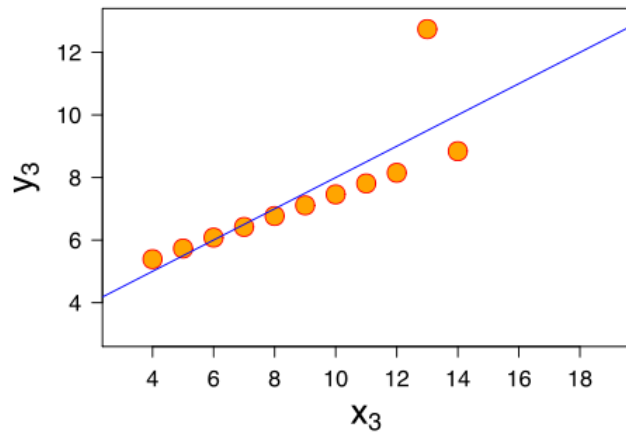
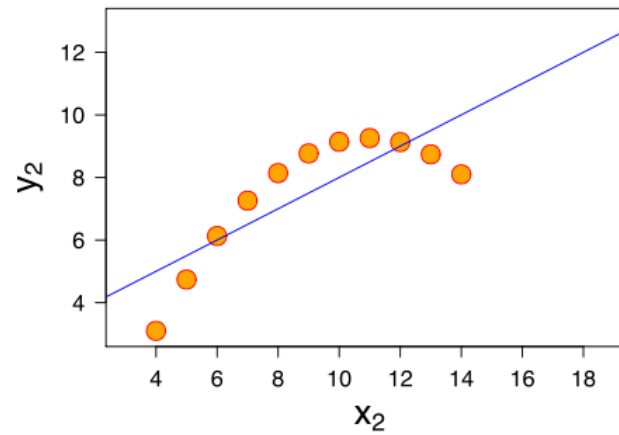
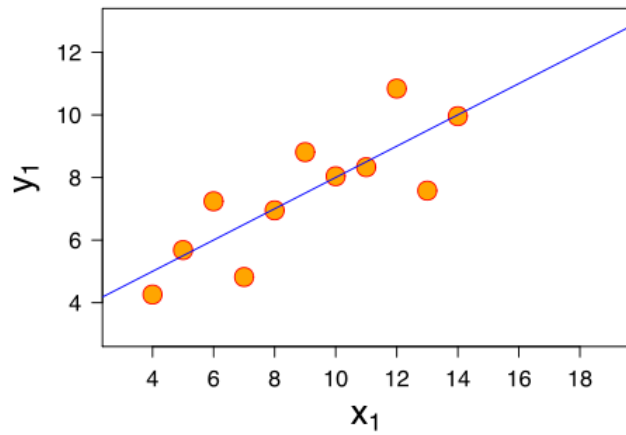
Why not use statistics

Anscombe's Quartet

- Mean, variance, correlation and linear regression line are all identical

I		II		III		IV	
x	y	x	y	x	y	x	y
10	8.04	10	9.14	10	7.46	8	6.58
8	6.95	8	8.14	8	6.77	8	5.76
13	7.58	13	8.74	13	12.74	8	7.71
9	8.81	9	8.77	9	7.11	8	8.84
11	8.33	11	9.26	11	7.81	8	8.47
14	9.96	14	8.10	14	8.84	8	7.04
6	7.24	6	6.13	6	6.08	8	5.25
4	4.26	4	3.10	4	5.39	19	12.5
12	10.84	12	9.13	12	8.15	8	5.56
7	4.82	7	7.26	7	6.42	8	7.91
5	5.68	5	4.74	5	5.73	8	6.89

Anscombe's Quartet



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Why use visualisations?

Abortion Rates By Age in the US

	1980	1985	1986	1987	1988	1989	1990	1991	1992	1993	1994	1995	1996	1997	1998	1999	2000	2001	2002	2003
	300	297	294	288	286	275	280	275	275	274	266	259	259	255	251	246	245	245	242	238
Less than 15 years old	607	624	605	578	553	523	515	502	511	492	488	479	493	498	504	497	512	519	529	537
15 to 19 years old	451	462	457	449	444	418	403	379	370	364	353	347	350	346	341	337	339	341	339	337
20 to 24 years old	310	328	328	327	327	318	328	330	333	334	326	317	314	307	301	297	296	298	296	293
25 to 29 years old	213	219	219	216	218	213	224	224	228	230	227	224	228	226	224	221	220	219	215	211
30 to 34 years old	213	203	201	197	194	189	196	192	192	189	183	179	178	176	174	171	169	171	169	167
35 to 39 years old	317	280	277	265	254	244	249	241	239	234	226	219	215	208	203	200	195	195	190	186
40 years old and over	461	409	381	374	361	350	354	339	338	329	320	309	301	291	290	283	276	276	278	268

Source: L.B. Finer and S.K. Henshaw, "Estimates of U.S. abortion incidence in 2001-2003", New York: Guttmacher Institute, 2006, <<http://www.guttmacher.org/pubs/2006/>> and unpublished data from Guttmacher Institute.

Which group has the lowest rate?

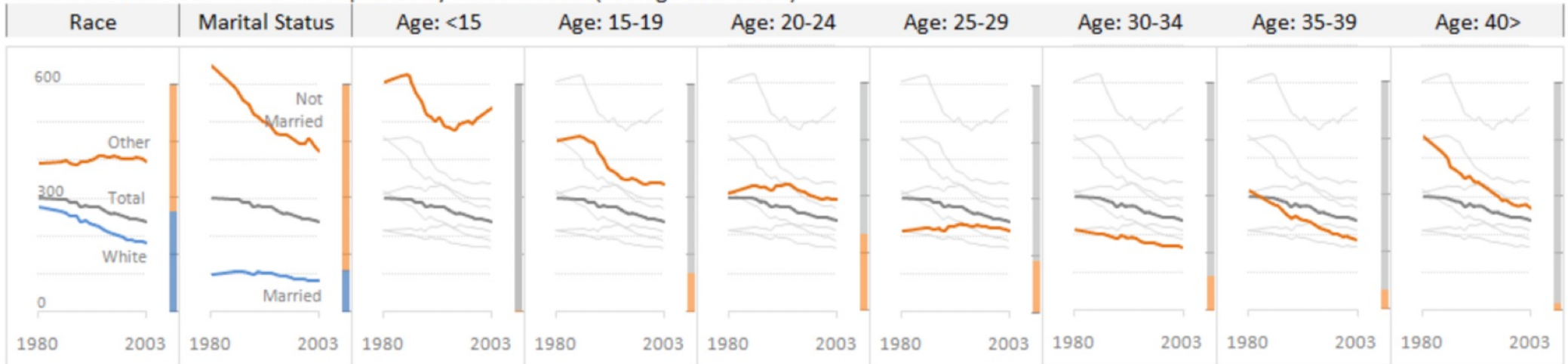
Are rates increasing or decreasing for any groups?

How fast/slow is the rate of change?

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Why use visualisations?

Abortion Ratios 1980-2003 and Proportion by Characteristic (average 2001-2003)



Which group has the lowest rate?

Are rates increasing or decreasing for any groups?

How fast/slow is the rate of change?

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[Jorge Camoes \(2008\) Chart Design: Abortion Ratios 1980-2003](#)

Early visualisations



Çatalhöyük town map, Turkey, c. 6200 BC

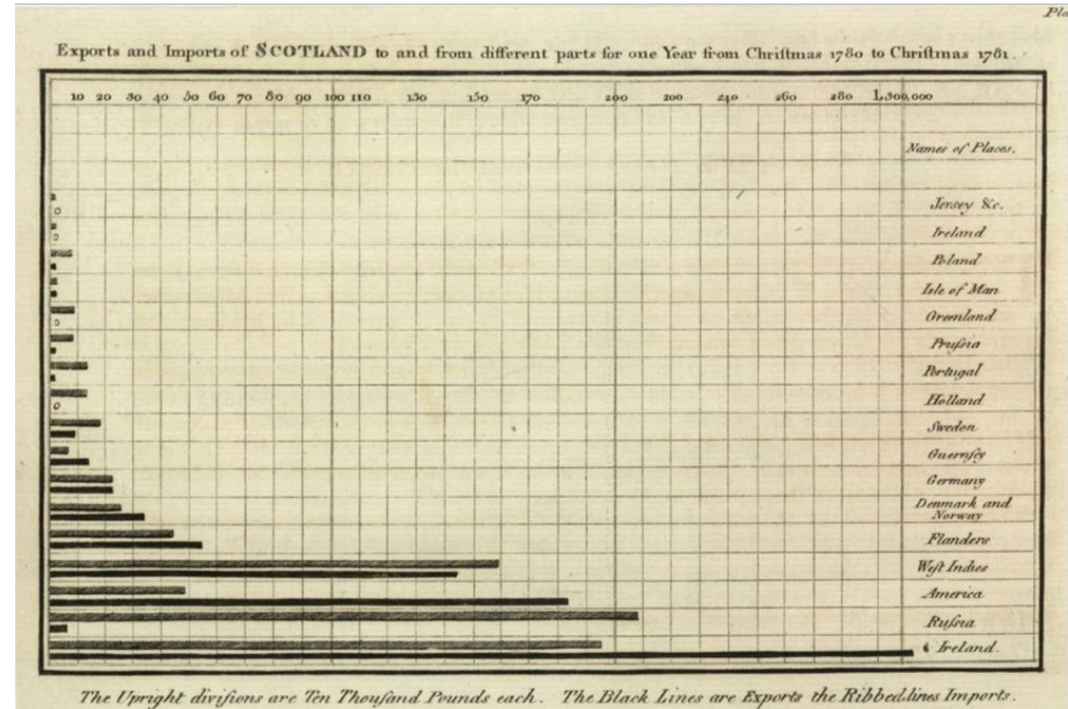
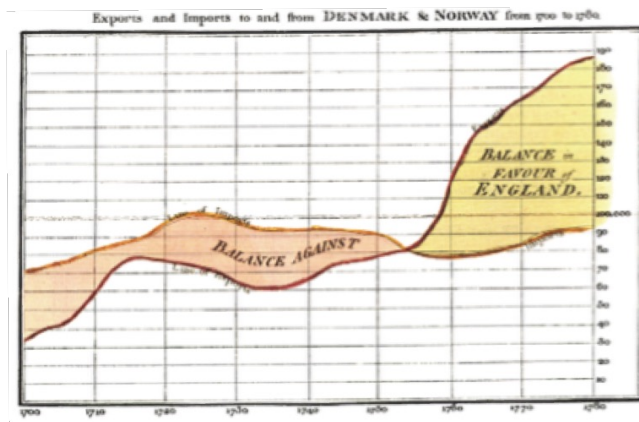
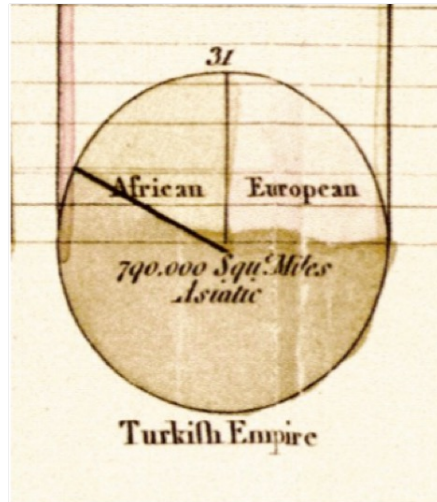
Sezer, A., (2021). Infographics designed in the field of tourism. Global Journal of Arts Education. 11(2), 141-147.

This historical map of the Atlantic Ocean and surrounding regions includes the following details:

- Geographical Labels:**
 - Seas and Oceans:** THE NORTH SEA, ATLANTICK OCEAN, THE ETHIOPICK OCEAN, SOUTH SEA.
 - Continents and Regions:** FLORIDA, PART OF THE PACIFIC, AMERICA, GUIANA, Gambia.
 - Islands and Archipelagos:** Bahama Isles, Canarie Isles, C. Verde, C. Verde, Calms and Tornado's, Azores, S. Johns.
 - Other Labels:** Bermuda, Cape, Spaguola, Caribbe Isles, C. Blanco.
- Wind Patterns:** Indicated by arrows and text labels such as "Variable Winds" and "Calms and Tornado's".
- Coordinates:** A grid system with latitude and longitude markings (e.g., 30, 45, 60, 70, 80, 90, 100, 110, 120, 130, 140, 150, 160, 170, 180, 190, 200, 210, 220, 230, 240, 250, 260, 270, 280, 290, 300, 310, 320, 330, 340, 350, 360).
- Map Features:** A compass rose is located in the center of the Atlantic Ocean, and a small inset map shows the location of the main map within the world's oceans.



Firsts...

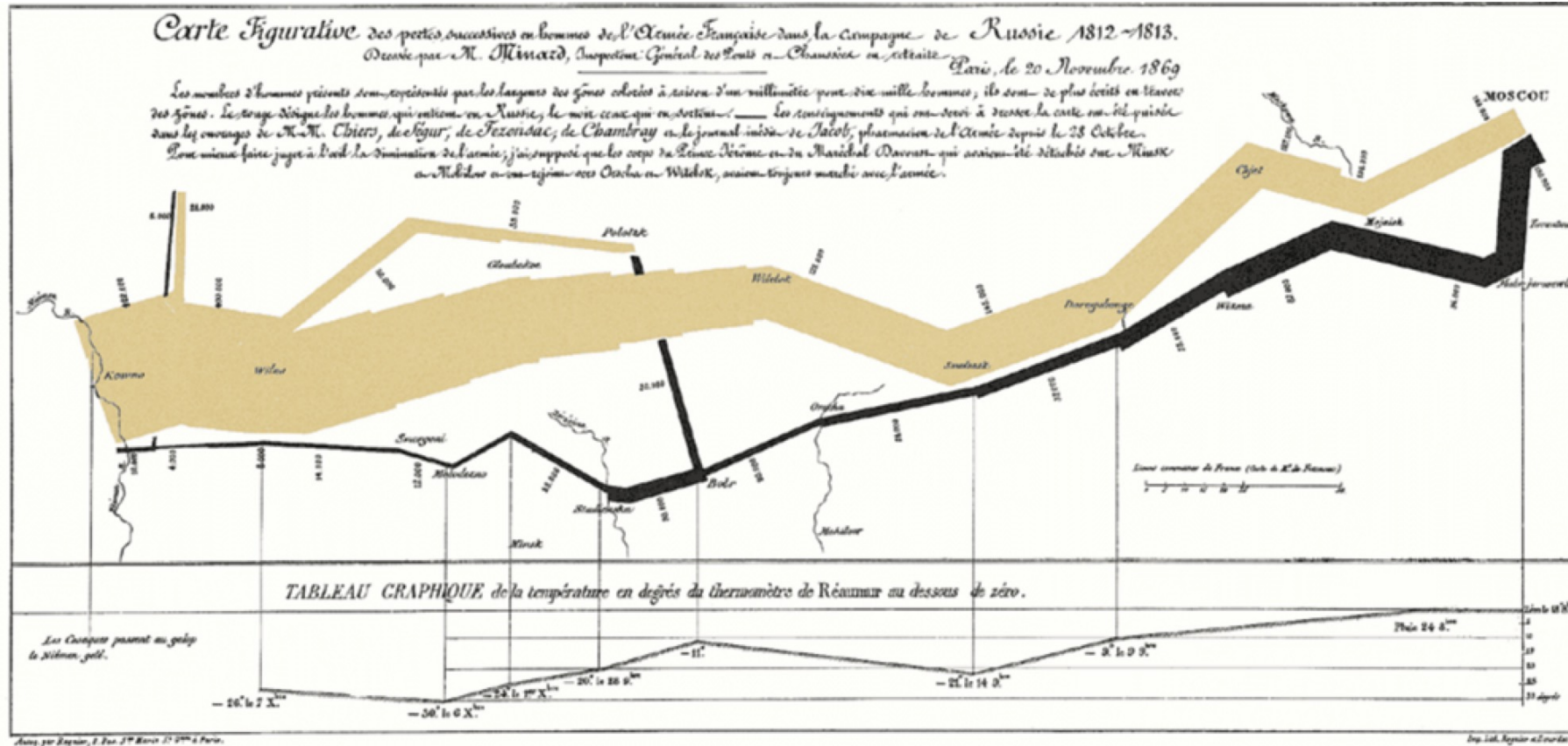


William Playfair

Scottish Engineer and
political economist, 1780's

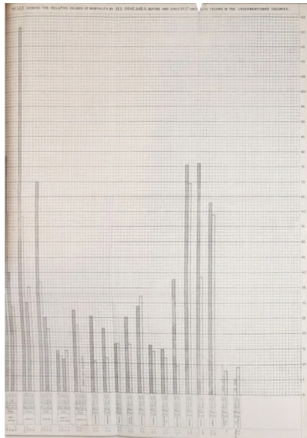


Classic early visualisations



Losses in the French army in the Russian campaign (1812-1813) by C. R. Minard (1869)

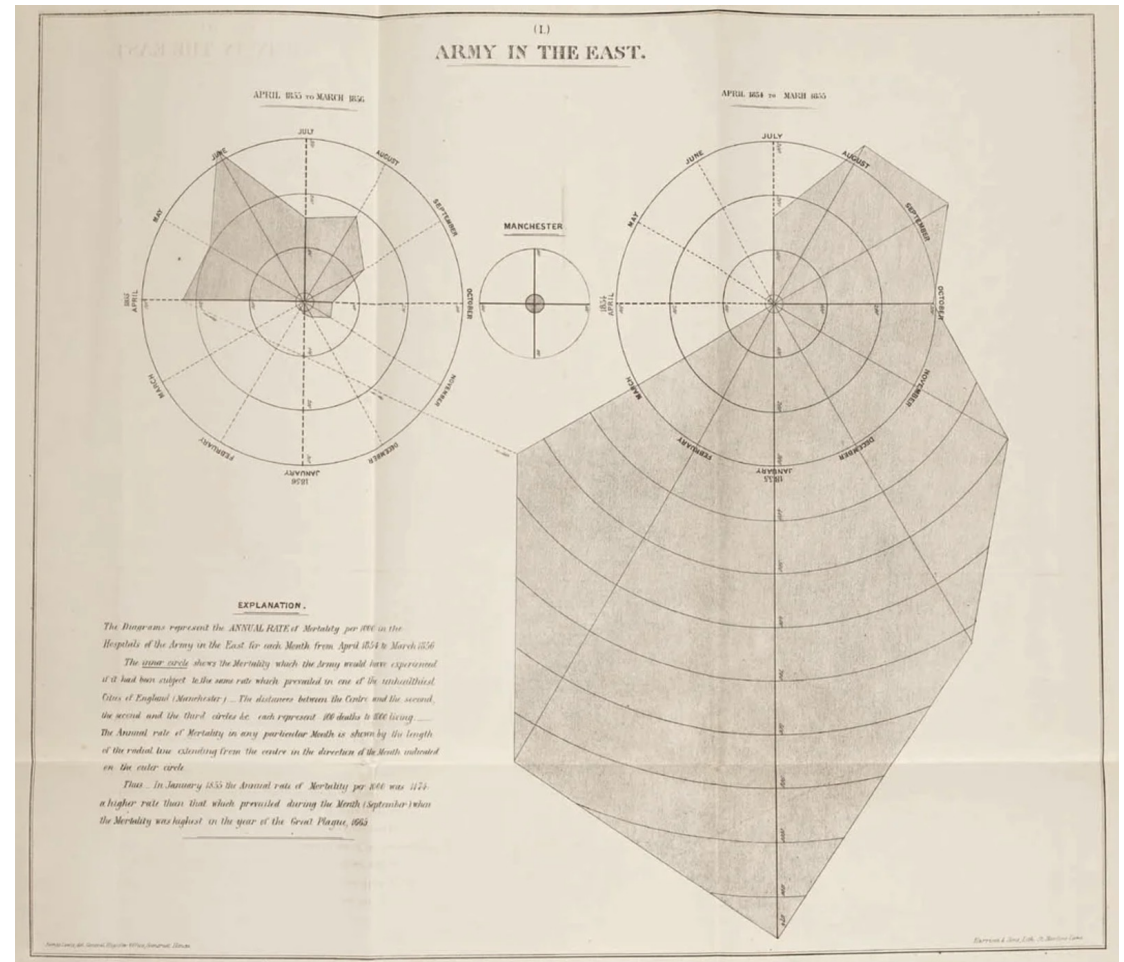
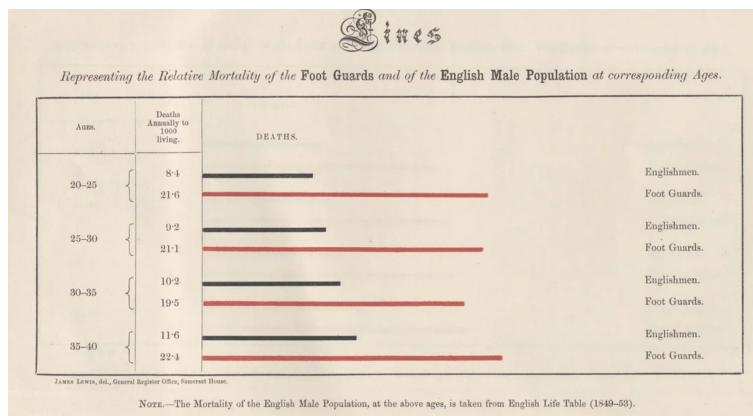
Early Classic visualisations



Pre-Nightingale chart of war deaths

Florence Nightingale

Nurse, social reformer
statistician, 1850s



<https://www.scientificamerican.com/article/how-florence-nightingale-changed-data-visualization-forever/>

Classic early visualisations

water pump

Broad Street water pump

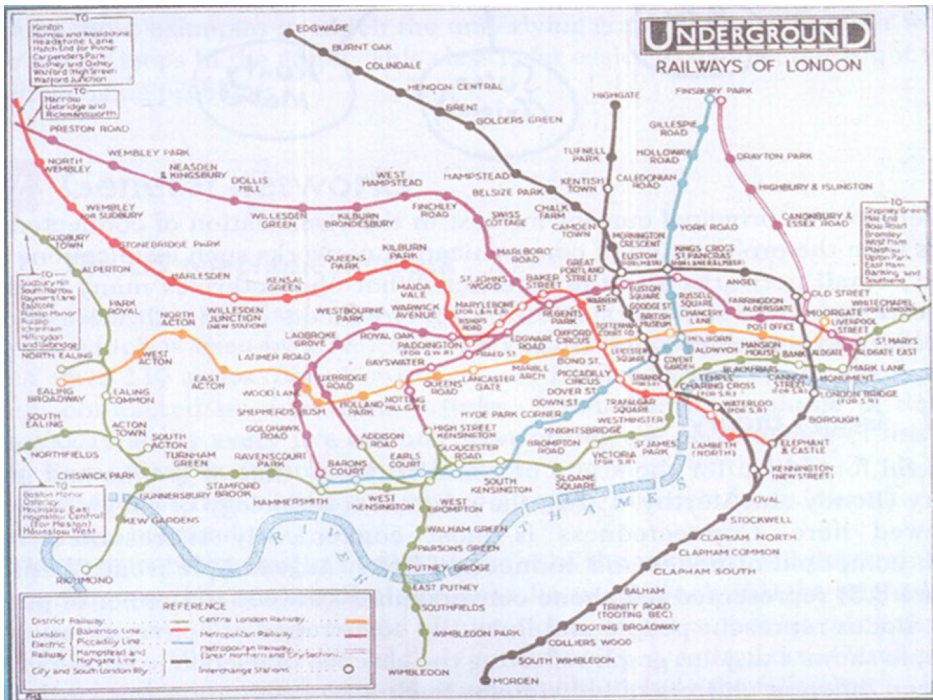
Cholera cases



JOHN SNOW
London physician and
pioneering
epidemiologist, 1850's



‘Modern’ classics...

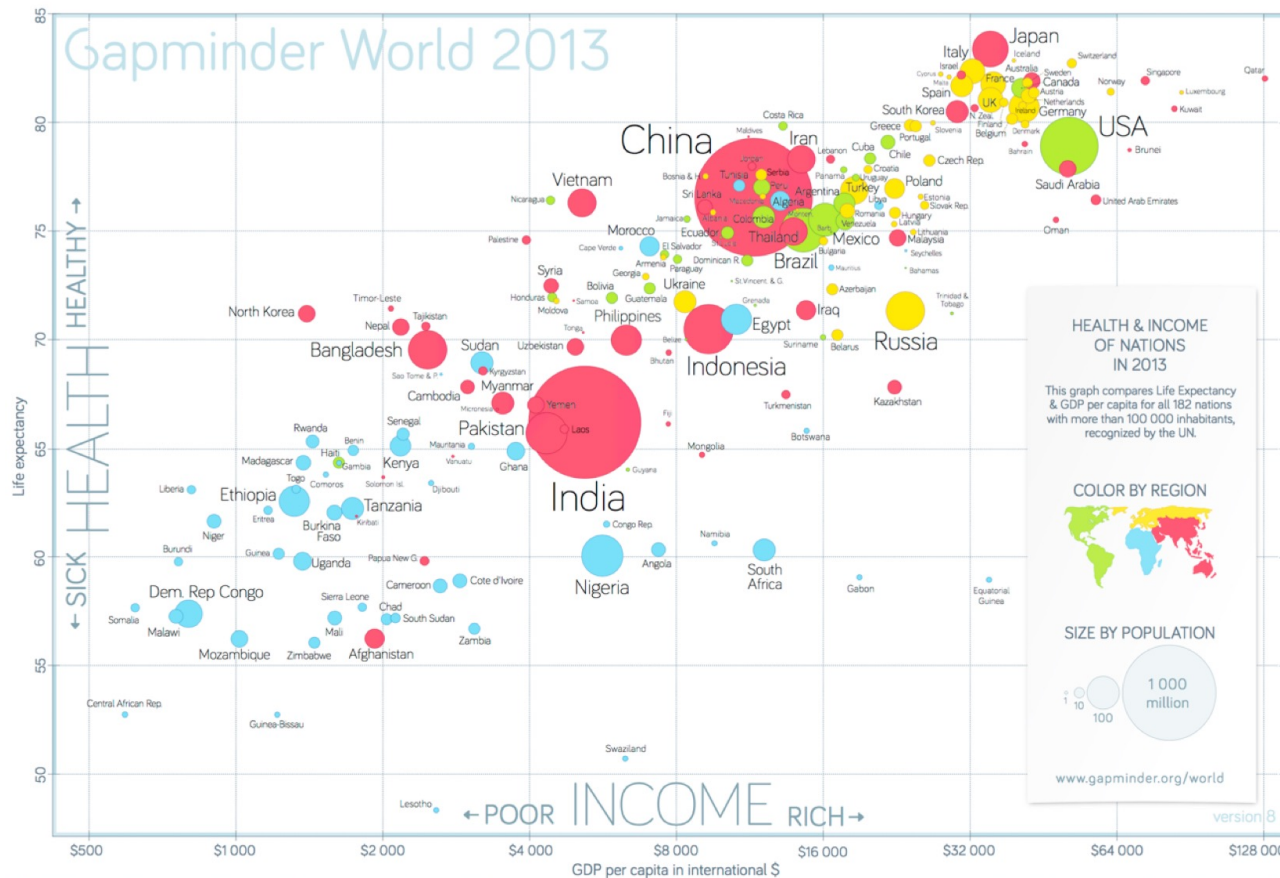


London Subway Map (1927)



Henry Beck's London Subway Map (1933)

Modern classics...



Watch the TED talk:

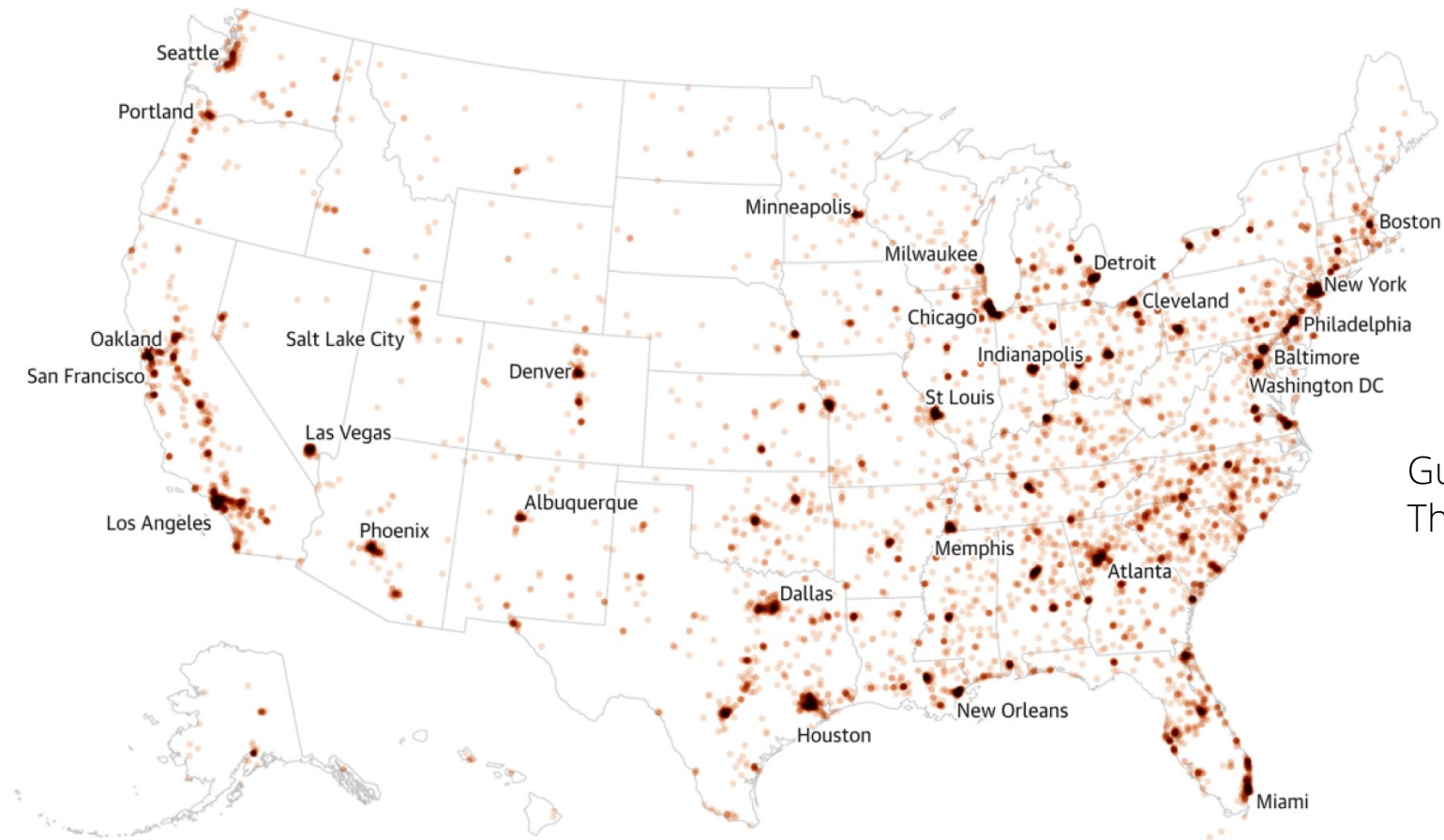
<https://www.youtube.com/watch?v=hVimVzgtD6w>

up to 6:18



HANS ROSLING
Swedish physician and
statistician

Modern storytelling



Gun Murders in USA
The Guardian

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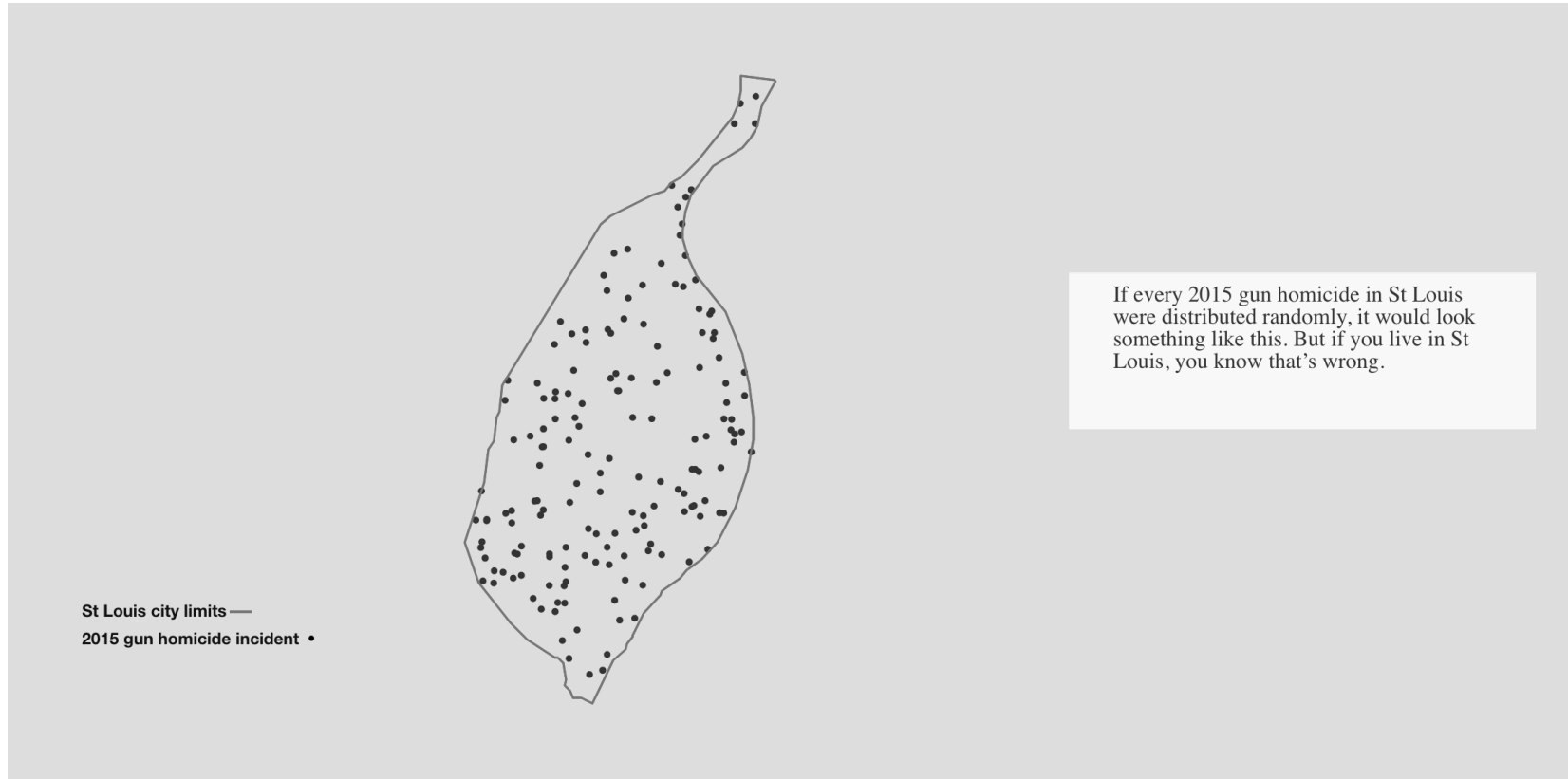
Modern story telling



North St Louis is home to blocks of abandoned and dilapidated homes. Houses like this one, where the rear wall has collapsed, are known locally as 'dollhouses'. Photograph: Sid Hastings for the Guardian

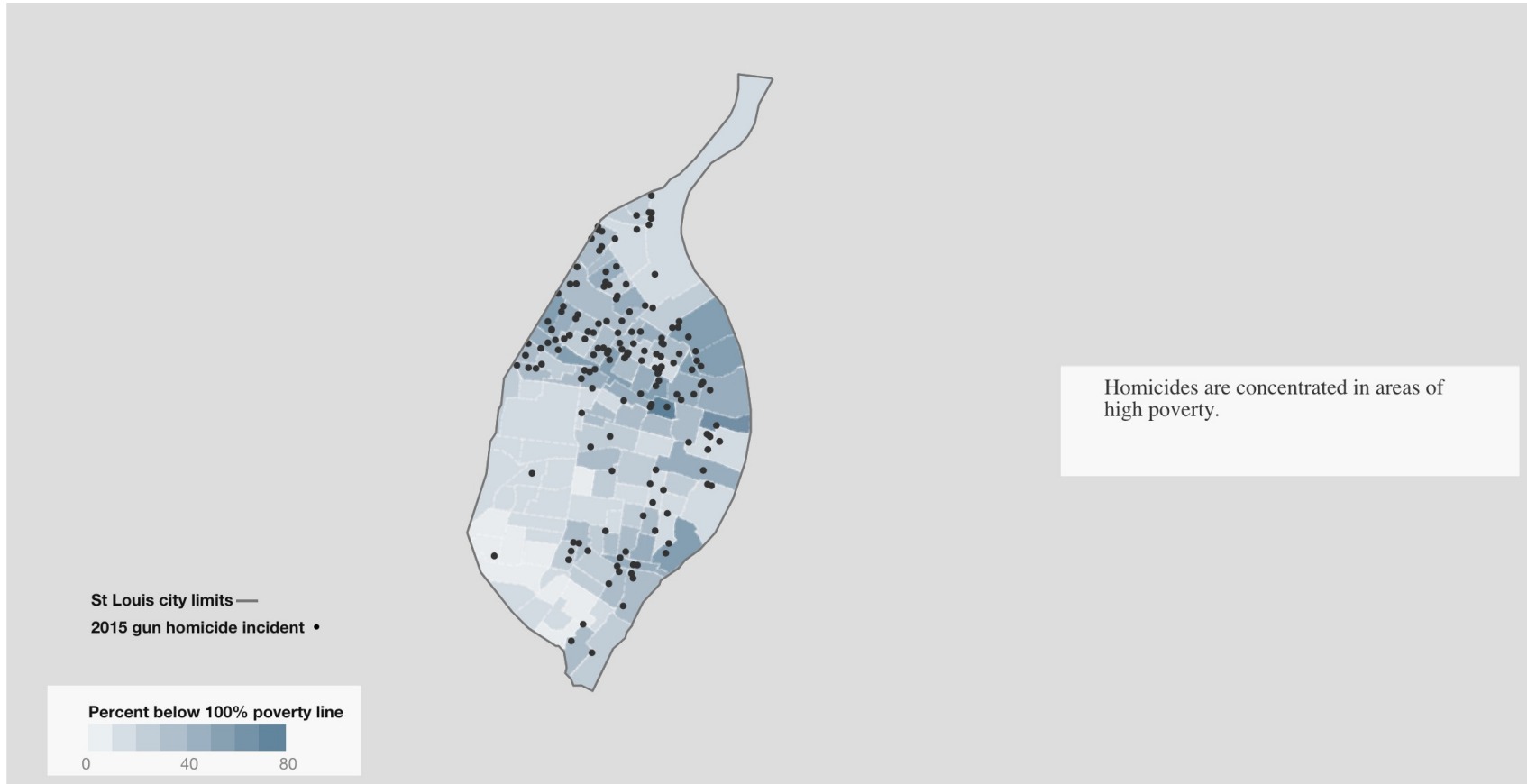
<https://www.theguardian.com/us-news/ng-interactive/2017/jan/09/special-report-fixing-gun-violence-in-america>

Modern storytelling



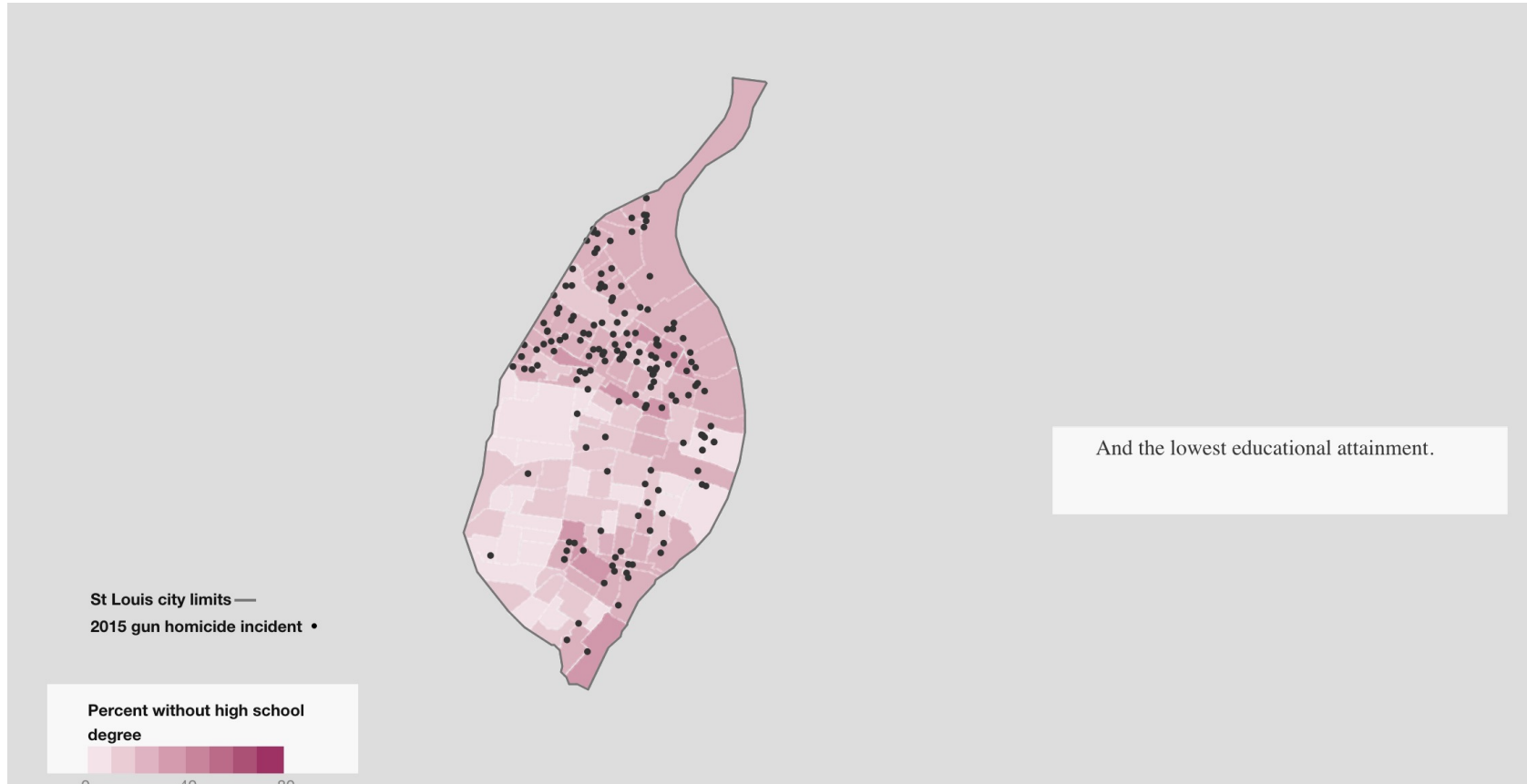
<https://www.theguardian.com/us-news/ng-interactive/2017/jan/09/special-report-fixing-gun-violence-in-america>

Modern storytelling



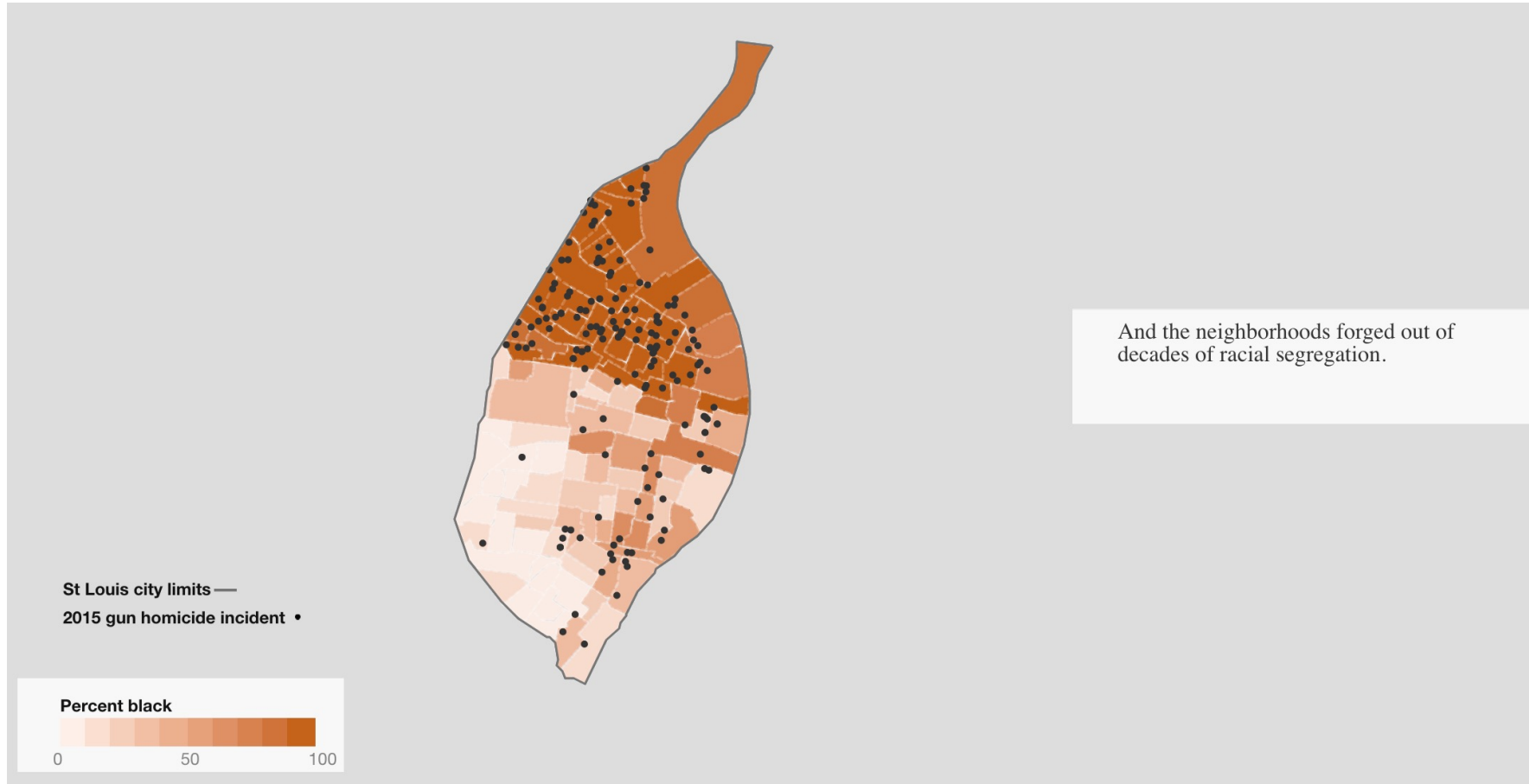
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Modern storytelling



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Modern storytelling

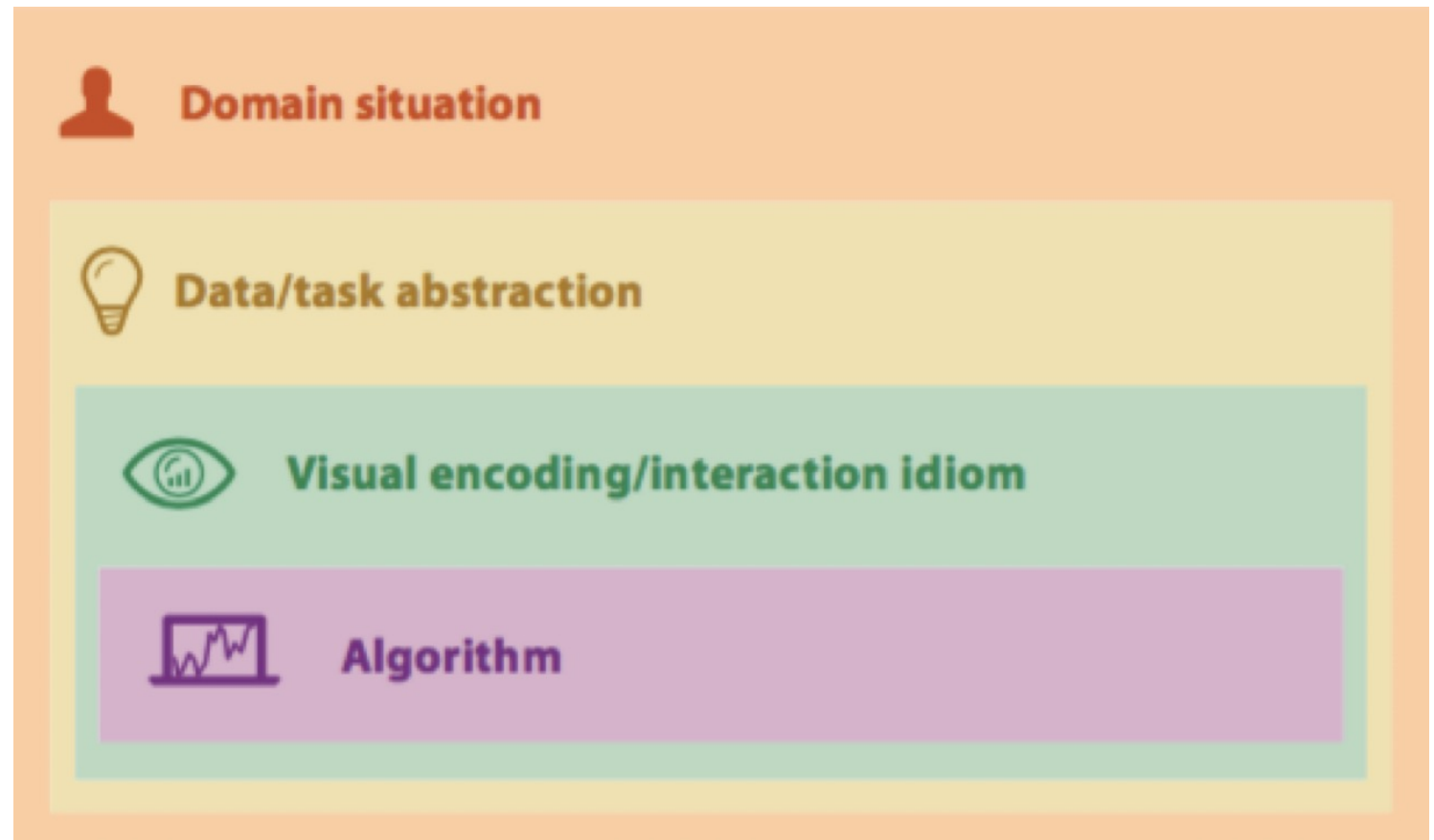


And the neighborhoods forged out of
decades of racial segregation.

<https://www.theguardian.com/us-news/ng-interactive/2017/jan/09/special-report-fixing-gun-violence-in-america>

Munzner Levels of Design

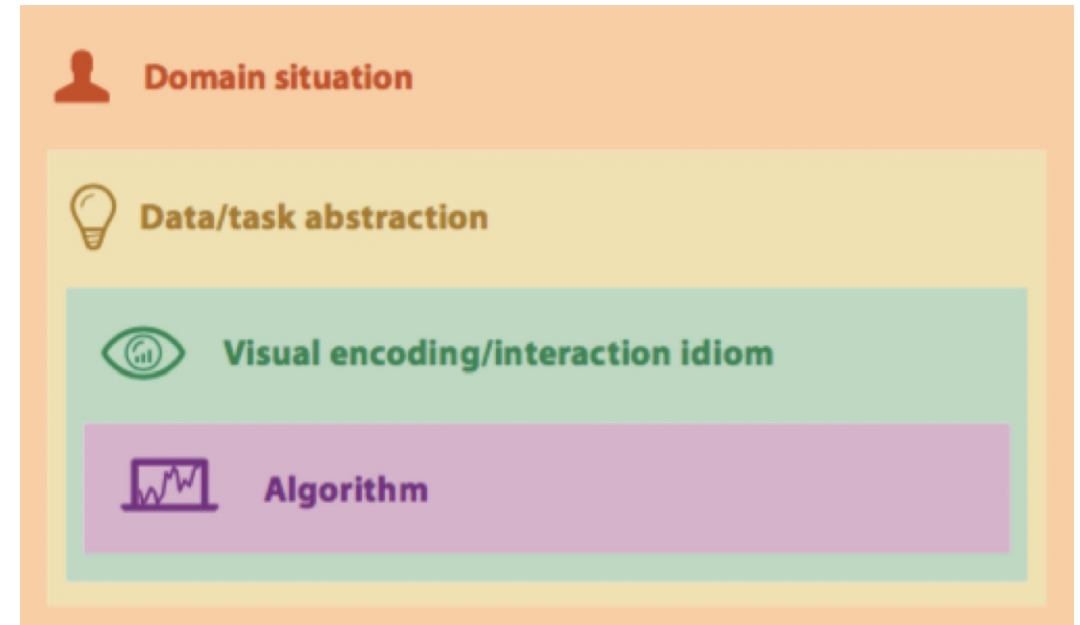
- Nested levels of design
- Wrong choices at one level can cascade to other levels



Domain situation

storytelling

- The users, their interest, their questions and their data
- Specialised vocabulary
- Learn about domain:
 - interviews, observation, literature review...
- Identify requirements

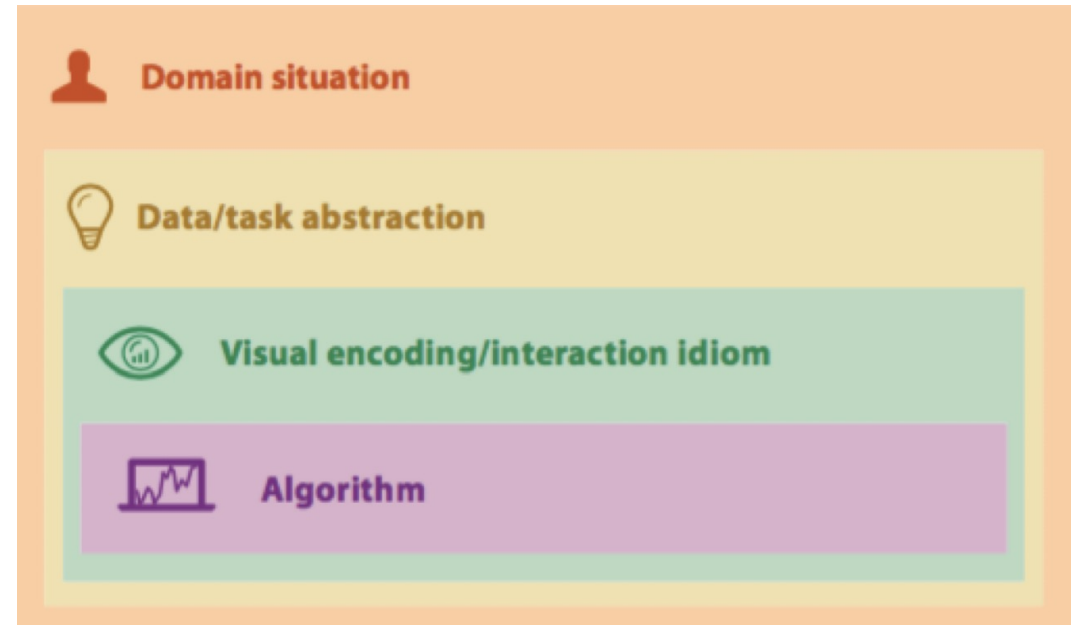


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Data/Task abstraction

Data processing

- Identify task and data types
- Try to identify domain independent task
 - “search for target in unknown location...”, “identify a trend...”, “find a correlation...”
 - Work out what data to use and how
 - encoding, transformation...

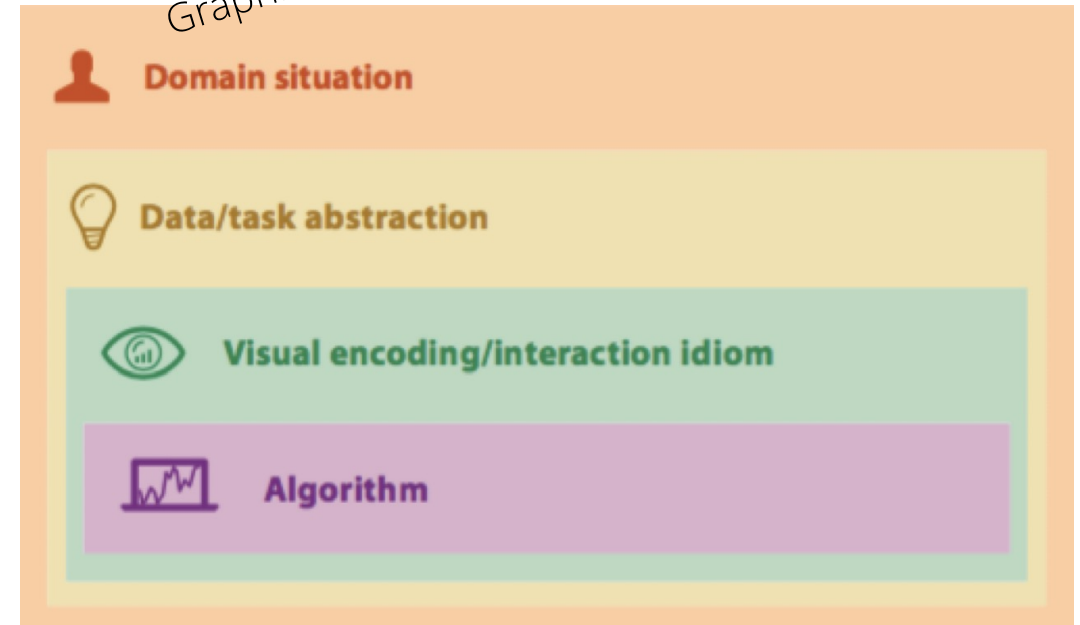


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Visual encoding/interaction idiom

Chart types
Graphical points

- Determine the visual representation of the data
 - idiom: a distinct visual approach
- Visual encoding idiom
- Interaction idiom
 - how users can change what they see



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Algorithm

- Procedure that allows a computer to implement visualisation

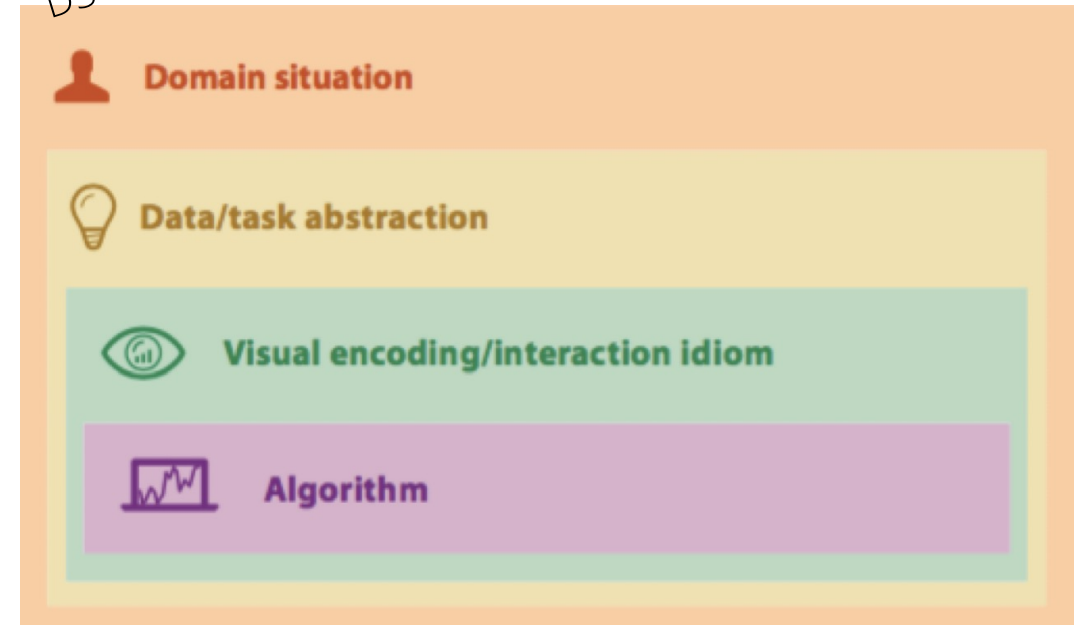
```
// Select the SVG container and set up dimensions
const svg = d3.select("svg")
  .attr("width", width)
  .attr("height", height);

// Set the scaling for the chart
const xScale = d3.scaleBand()
  .domain(d3.range(data.length))
  .range([0, width])
  .padding(0.1);

const yScale = d3.scaleLinear()
  .domain([0, d3.max(data)])
  .range([0, height]);

// Create the bars for the chart
svg.selectAll("rect")
  .data(data)
  .enter()
  .append("rect")
  .attr("x", (d, i) => xScale(i))
  .attr("y", d => height - yScale(d))
  .attr("width", xScale.bandwidth())
  .attr("height", d => yScale(d))
  .attr("fill", "steelblue");
```

D3 code



Munzner, T. (2014). *Visualization analysis and design*. CRC press

AI declaration: code generated by Github Copilot

Visualization is really about external cognition, that is, how resources outside the mind can be used to boost the cognitive capabilities of the mind.



Stuart Card

Pioneering Human-
Computer Interaction
(HCI) researcher

